

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

December 2011

EE 206: DIGITAL ELECTRONICS & MICROPROCESSOR

Date: 09.12.2011, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Why are NAND and NOR gates called universal gates? [04]

(b) Explain De Morgan's theorem. [03]

Q - 2 (a) Fill in the blanks and multiple choice questions : [05]

(i) $A' + B' + C' + D' = D$ represents a _____ gate.

(I) NAND (II) NOR (III) OR

(ii) $AB' + A'B = D$ represents _____ gate.

(I) OR (II) EX-OR (III) EX-NOR

(iii) The expression $Y = AB + AB' + A'B'$ is in the _____ form.

(I) Standard SOP (II) Standard POS.

(iv) The minterm corresponding 011 combination is _____

(I) $A'BC$ (II) $A' + B + C$ (III) $A + B' + C$

(v) A byte is a combination of _____ binary bits.

(I) 4 (II) 8 (III) 16

(b) Prove that [05]

(i) $(A+C)(A+D)(B+C)(B+D) = AB + CD$

(ii) $(B+BC)(B+B'C)(B+D) = B$

(c) Do as directed : [04]

(i) Obtain the 2's complement of $(10110010)_2$

(ii) Convert $(6AC)_{16} = (\quad)_{10} = (\quad)_2$

OR

Q - 2 (a) Explain 3-bit Asynchronous up counter with timing diagram. [05]

(b) Explain with diagram BCD to 7-segment decoder using 4 line to 16 line decoder. [05]

(c) What is the function of a shift register? Give its applications. [04]

- Q - 3 (a) Minimize the following Boolean expression using K-map and realize it using the basic gates. $Y = \sum m(1,3,5,9,11,13)$. [05]
- (b) What is meant by multiplexer? Write the advantages of multiplexers and Obtain an 8:1 multiplexer using two 4:1 multiplexers. [05]
- (c) Compare between TTL and CMOS logic [04]

OR

- Q - 3 (a) Explain with diagram the working of T-type flip flop. Give its truth table. [05]
- (b) Simplify the expression given below using K-map. The don't care conditions are indicated by d().
 $Y = \sum m(1,3,7,11,15) + d(0,2,5)$ [05]
- (c) Explain the working of a half adder? Draw its logic diagram. [04]

SECTION – II

- Q - 4 (a) Explain parity flag and auxiliary flag in 8085 microprocessor. [04]
- (b) Explain the following term [03]
- (i) Instruction cycle (ii) Machine cycle (iii) T-state.

- Q - 5 (a) Draw pin diagram of 8085 and give function of ALE, SOD and trap pin. [07]
- (b) Explain addressing modes of 8085. [07]

OR

- Q - 5 (a) Draw architecture of 8085 microprocessor and explain ALU. [07]
- (b) Write a short note on memory classification. [07]

- Q - 6 (a) Write 8085 microprocessor based assembly program of the following. Explain the program logic in brief. [09]
- (i) Add 16 bit no. in memory location 40 & 41 to 42 & 43. The 8-bit m.s.b. of the two no. to be added are in memory location 41 & 43. Store the result in memory location 44 & 45 with m.s.b.
- (ii) Write a program to shift an eight bit data four bits right. Assume that data is in register C.
- (iii) Load the hexadecimal numbers 9B H and A7 H in register D and E, respectively and add the numbers. If sum is greater than FF H, display 01 H at output0; otherwise display the sum.

- (b) Write similarities and differences between the instructions CALL and RET & PUSH and POP instructions. [05]

OR

- Q - 6 (a) Find out the total number of zeros, positive and negative numbers from the data array stored in memory. Store the result in memory locations away from each other by 10 decimal count. [08]

- (b) Explain the function of following instructions. [06]
- (i) LDA addr. (ii) SUB M (iii) CMA
 (iv) RAL (v) XRI data (vi) INR M